0/12 Questions Answered

Lab #06 - Written part

Q1 Attendance/engagement

40 Points

Empty placeholder for us. No need to write anything here.

Save Answer

Q2 The table in memory
8 Points

Q2.1 Visiting order table
5 Points

Export your document containing the visiting order table as a **PDF file**, and upload it here.

Please select file(s) Select file(s)

Save Answer

Q2.2 Fetches in the tiny table
1 Point

In the tiny table, how many fetches does each function need in total?

Save Answer

Q2.3 Bytes between consecutive visit
2 Points

In the real table (the one with 100,000 drivers), each row holds 168 `int`s.

- How many bytes apart in memory are two values visited one after the other by `TotalsHourByHour()`?
- And by `TotalsDriverByDriver()`?

Save Answer

Q3 Driver by driver

1 Point

- Which function do you expect to be faster, `TotalsHourByHour()` or `TotalsDriverByDriver()`?
- By roughly how many times?
- Explain your reasoning in one or two sentences.

This prediction is graded on **completion only, not on correctness**.

Save Answer

Q4 Optimizing
5 Points

Q4.1 Measured times
1 Point

Report a typical time for both functions, without and with `-O2`.

Save Answer

Q4.2 Speed-up ratios
2 Points

- Without `-O2`, how many times faster is `TotalsDriverByDriver()` than `TotalsHourByHour()`?
- And with `-O2`?

Save Answer

Q4.3 Prediction check and explanation
2 Points

Was your prediction of step 3 right? Use your fetch counts of step 2 to explain why `TotalsDriverByDriver()` is faster.

Save Answer

Q5 Planning the split
6 Points

Q5.1 Prediction
1 Point

In this part, function `TotalsByMetro()` will compute the six chunks sequentially (one per metro area), on a single core.

- Do you expect it to be faster than `TotalsDriverByDriver()` of Part #1, slower, or about the same?
- Briefly explain why.

This prediction is graded on **completion only, not on correctness**.

Save Answer

Q5.2 Load imbalance
2 Points

Suppose now that the six chunks were computed in parallel, each by its own core, and that the time of each chunk was proportional to its number of drivers.

- Which metro area would finish last?
- At best, how many times faster than computing them sequentially would `TotalsByMetro()` then be?

Save Answer

Q5.3 Step that cannot be split
1 Point

Suppose that loading the data (that is, generating the drivers' data in this lab) takes 100 ms and cannot be split, and that computing the totals takes 5 ms on a single core.

- If the totals were split into 16 equal chunks, computed in parallel by 16 cores, about how long would the whole program take?

Save Answer

Q5.4 Splitting by day
2 Points

A classmate suggests another split, with 7 chunks computed in parallel, one per day of the week: each core would compute the totals of its 24 hours, by visiting all the drivers for each of its hours.

- Which function of Part #1 would the work of each core resemble?
- Would each chunk be processed quickly?

Save Answer

Save All Answers

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